	ZEAL EDUCATION SOCIETY'S **ZEAL COLLEGE OF ENGINEERING AND RESEARCH** **NARHE │PUNE -41 │ INDIA**	
Record No.: **ZCOER-ACAD/R/16M**	Revision: **00**	Date:**01/04/2021**

Question Bank

Department: Semester: II **Academic Year:** 2024-2025

Class: F.Y.B.Tech.**Div:** **Date:**

Course: Engineering Mathematics II

Unit No. –5 name	Q. No.	Question	Marks	CO	Blooms Level
Multiple Integral	1.	Evaluate $\int_0^1 \int_0^x e^{\left(\frac{y}{x}\right)} \, dxdy$.	05	CO5	2
	2.	Evaluate $\int_1^{\ln 8} \int_0^{\ln y} e^{x+y} \, dx \, dy$	05	CO5	2
	3.	Evaluate $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dydx}{1+x^2+y^2}$	05	CO5	2
	4.	Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{dydx}{(1+e^y)(\sqrt{1-x^2-y^2})}$	05	CO5	2
	5.	Evaluate $\iint y \, dx \, dy$ over the region enclosed by the $x^2 = y$ and the line $y = x + 2$.	05	CO5	3
	6.	Evaluate $\iint xy \, dx \, dy$ over the region enclosed by the $x^2 = y$ and the line $y^2 = -x$.		CO5	2
	7.	Change the order of integration and evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} \, dxdy$	05	CO5	2
	8.	Change the order of integration and evaluate $\int_0^\pi \int_x^\pi \frac{siny}{y} dx \, dy$	05	CO5	2
	9.	By changing order of integration evaluate $I = \int_0^1 \int_{e^x}^e \frac{dydx}{logy}$	05	CO5	2

	10	Evaluate $\iint \frac{x^2y^2}{x^2+y^2}$ dx dy where R is region bounded by the curves $x^2 + y^2 = 4 \ \ and \ x^2 + y^2 = 9$			
	11	Evaluate $\iint x^2y^2 \ dx \ dy$ over the positive quadrant $x^2 + y^2 = a^2$ of the circle using polar transformation.	05	CO5	2
	12	Evaluate $\iiint \frac{dx \ dy \ dz}{\sqrt{1-x^2-y^2-z^2}}$ taken through the volume of sphere $x^2 + y^2 + z^2 = 1$ in positive octant.	05	CO5	3

Course faculty

Open Book Test

Department: **Semester:** II **Academic Year:** 2024-2025

Class: F.Y.B.Tech.**Div:** **Date:**

Course: Engineering Mathematics II

Unit No. –5 name	Q. No.	Question	Marks	CO	Blooms Level
Multiple Integral	1.	Evaluate $\int_0^1 \int_0^x e^{\left(\frac{y}{x}\right)}\ dxdy$.	05	CO5	2
	2.	Evaluate $\iint y\ dx\ dy$ over the region enclosed by the $x^2 = y$ and the line $y = x + 2$.	05	CO5	2
	3.	Change the order of integration and evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y}\ dxdy$	05	CO5	2

OR

Unit No. –5 name	Q. No.	Question	Marks	CO	Blooms Level
Multiple Integral	1.	Change the order of integration and evaluate $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dx\ dy$	05	CO5	2
	2.	Evaluate $\iiint \frac{dx\ dy\ dz}{\sqrt{1-x^2-y^2-z^2}}$ taken through the volume of sphere $x^2 + y^2 + z^2 = 1$ in positive octant.	05	CO5	2

	3.	Evaluate $\iint x^2y^2\,dx\,dy$ over the positive quadrant $x^2+y^2=a^2$ of the circle using polar transformation.	05	CO5	2